Lesson notes · for you

Let the computer search: unit check

Optimization · Level 3 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Try them all, or think ahead
- Shortest paths
- The bus route
- Timetables with CP-SAT

Activity: Activity: Search race

1. Warm-up (3 min): For each lesson, write one search from your life that it could solve: packing, a way to school, a round trip, a timetable.
2. Race (10 min): With a partner, pick one small problem. One of you solves it with a quick rule (greedy or nearest neighbour), the other with a sure method (every option, Dijkstra, or a small model).
3. Compare (4 min): Compare your answers and how long each took. When did the quick rule miss?
4. Fair (3 min): Look at the best answer and ask who it would be hard on: the first stop, the last slot, the school that got nothing.

Keep the problems small (four or five items, places or tasks) so both methods finish in the time.

Discussion

- When is a quick rule good enough?
- Which method surprised you most?
- Who should decide what “fair” means in a search?

Check yourself

- The method I would trust for my own search + one fairness question I would ask.

Try it this week

Pick one search from your own life (packing a bag, a route you take, a week to plan). Solve it with code or on paper using a method from this unit, then check whether the answer is fair to everyone it affects.

Worksheet

Let the computer search: unit check

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Name _____ Class _____ Date _____

1 **A brute-force search over 10 items checks 1,024 possible loads.**

◯ True ◯ False

2 **What does this program print?**

```
w = [3, 2, 2]
v = [5, 3, 3]
best = 0
for mask in range(2 ** 3):
    kg = sum(w[i] for i in range(3) if mask >> i & 1)
    pts = sum(v[i] for i in range(3) if mask >> i & 1)
    if kg <= 4 and pts > best:
        best = pts
print(best)
```

◯ 6 ◯ 5 ◯ 11

3 **What does Dijkstra's method need to give the right answer? Choose all that apply.**

☐ No road with a negative length ☐ Settling the smallest waiting number first

☐ Places listed in alphabetical order ☐ Every road the same length

4 **This Dijkstra program finds the longest way instead of the shortest. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  while todo:
2      p = min(todo, key=lambda q: dist[q])
3      todo.remove(p)
4      for q, km in roads[p].items():
5          dist[q] = max(dist[q], dist[p] + km)
```

5 **Put in order: solving a timetable with CP-SAT.**

☐ Check the result with the people involved

☐ Read the status

☐ Run the solver

☐ Write each rule as a line

☐ Give each choice a variable with its domain